

Vaishnav Sreekumar

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SUMMARY

Computer Science undergraduate specializing in AI-powered backend systems, cloud platforms, and distributed software engineering. Experienced in building production-style applications using Python, AWS, LLMs, and Go with hands-on open-source development.

SKILLS

Languages: Python, Go, Java, SQL, JavaScript

AI: LLMs, OpenAI API, Prompt Engineering, Pydantic

Backend: REST APIs, Distributed Systems, Microservices

Cloud: AWS (Lambda, EventBridge, DynamoDB, IAM, CloudWatch), Terraform, Docker, Kubernetes

Data & Observability: PostgreSQL, Redis, Kafka, Prometheus, Grafana, ELK

Tools: Linux, Git, GitHub

OPEN SOURCE CONTRIBUTIONS

- **Kerno (GSSoC 2026)** Go | Cloud Native | Linux
Contributed production-quality features to an open-source cloud observability platform by implementing the **kerno doctor -list-rules** CLI capability, designing automated unit tests, improving rule discovery workflows, and collaborating with maintainers through iterative code reviews, CI validation, and pull request discussions.
- **Kubernetes Ecosystem – kubernetes-network-drivers (#24)**
Contributed improvements to Kubernetes Dynamic Resource Allocation (DRA) authorization workflows, strengthening RBAC-based access control for cloud-native infrastructure and gaining experience working with large-scale upstream open-source development.

SELECTED PROJECTS

- **SentinelFinOps: AI-Powered Cloud Governance & FinOps Platform** GitHub
Python OpenAI API AWS Terraform Lambda DynamoDB EventBridge Pydantic
 - Architected an enterprise-grade AI-powered cloud governance platform comprising **35+ modules**, integrating LLM reasoning, deterministic policy validation, and AWS automation to generate explainable infrastructure optimization recommendations.
 - Engineered a modular AI platform with **9 AI subsystems** including Context Mapping, Prompt Registry, AI Gateway, Runtime Orchestration, Schema Validation, Policy Engine, Telemetry, and Evaluation Framework supporting provider-agnostic execution.
 - Built multi-account AWS automation using Organizations, Lambda, EventBridge, CloudWatch, IAM, DynamoDB, and Terraform with distributed remediation using DynamoDB Conditional Writes to eliminate concurrent execution race conditions.
 - Delivered production-quality software backed by **116+ automated unit/integration tests**, immutable Pydantic contracts, dependency injection, runtime validation, and extensible AI provider support.
- **SentinelX: Multi-Tenant SIEM & Observability Platform** GitHub
Python Apache Kafka Prometheus Grafana ELK Stack Linux
 - Developed a multi-tenant SIEM platform using Kafka, Prometheus, Grafana, and ELK Stack for scalable real-time security telemetry ingestion, monitoring, and visualization.
 - Built detection pipelines for SSH brute-force attacks, reconnaissance, suspicious authentication activity, and malicious web requests with automated alert generation.
- **Zero Trust Access Platform** GitHub
Go PostgreSQL Docker JWT REST APIs
 - Built secure backend services in Go implementing REST APIs, JWT authentication, RBAC, MFA, and identity-aware authorization following Zero Trust principles.
 - Developed Dockerized microservices with PostgreSQL emphasizing secure coding, modular architecture, and scalable authentication workflows.

TECHNICAL WRITING

How I Prevented Race Conditions in AWS Lambda Using DynamoDB Conditional Writes Read Article
Published a deep-dive technical article explaining the design and implementation of distributed locking using DynamoDB Conditional Writes, optimistic concurrency control, and fault-tolerant AWS Lambda execution for production-grade cloud systems.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner (CLF-C02)** Amazon Web Services
- **IBM Machine Learning with Python** IBM | Coursera

EDUCATION

Vellore Institute of Technology, Bhopal Sep 2023 – Sep 2027
Bachelor of Technology (B.Tech), Computer Science and Engineering
– CGPA: **8.15 / 10**